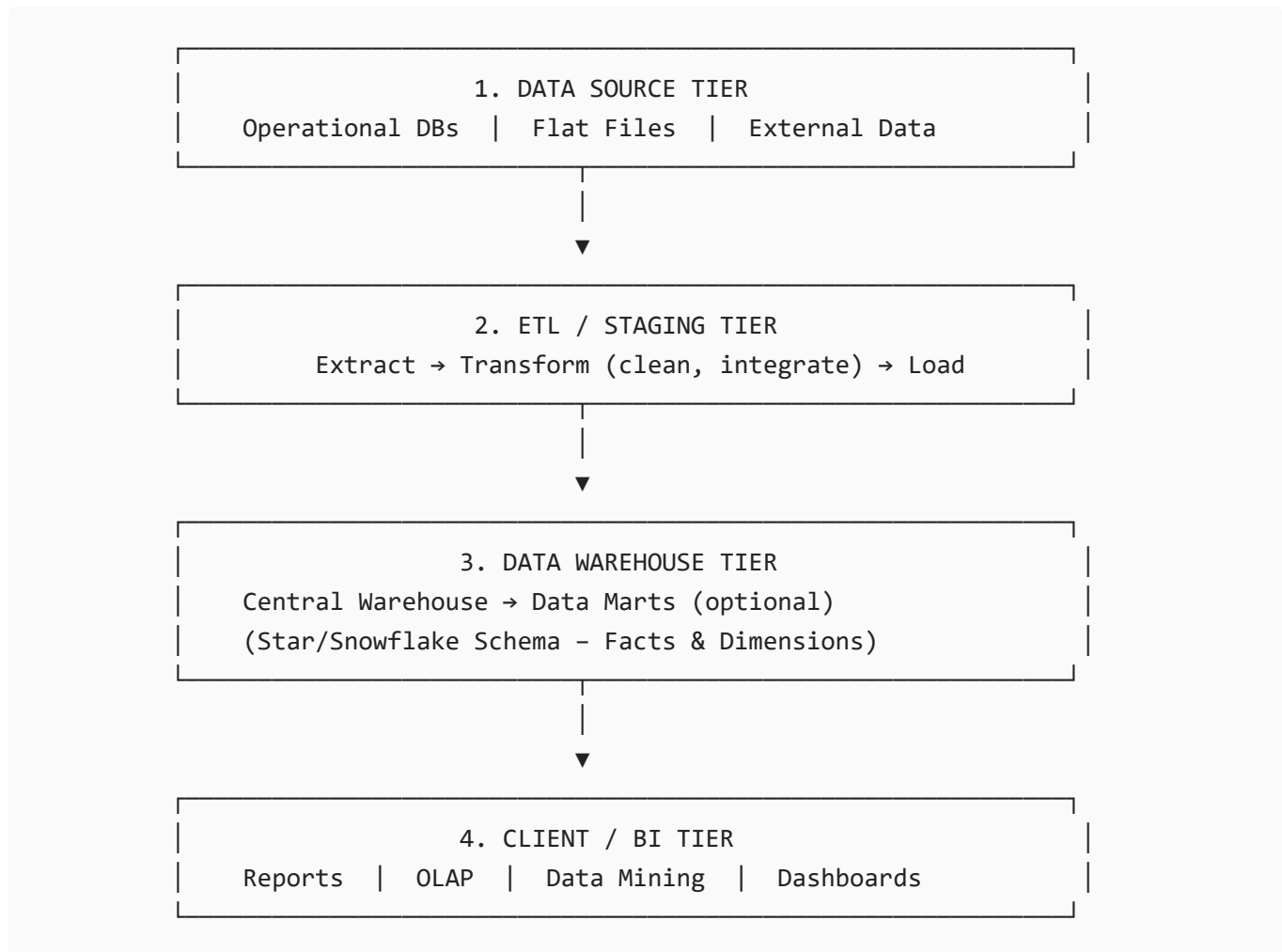


DMBI Module 2

Define data warehouse. Explain data warehouse architecture with a diagram.

A Data Warehouse is a centralized repository that stores integrated, historical, and non-volatile data collected from multiple heterogeneous sources, specifically structured for querying, reporting, and analysis rather than for daily transaction processing.

Data warehouse architecture:



1. DATA SOURCE TIER:

- Includes all source systems such as operational databases, flat files, and external feeds. This tier provides raw, unprocessed data needed for business analysis.

2. ETL / STAGING TIER:

- Extracts data from sources, transforms it by cleaning, validating, and integrating, then loads it into the warehouse. Prepares consistent, high-quality data.

3. DATA WAREHOUSE TIER:

- Central repository storing integrated, historical data using star or snowflake schemas. May include data marts for department-specific reporting and analysis.

4. CLIENT / BI TIER:

- The top layer where users access data via reports, OLAP cubes, dashboards, and mining tools. This tier delivers insights and decision support.
-

What is a Decision Support System?

A computer-based system that supports decision-making using data and models

- Helps in solving semi-structured and unstructured problems
- Provides what-if analysis and forecasting
- Supports managers in making informed decisions

Characteristics:

1. Interactive Interface:

- The graphical user interface is also user friendly hence users can interact with DSS easily in inputting their data and get the desired output.

2. Data Integration:

- It gets information from various sources like DBMS, data marts, Data warehouses and even data feeds to have complete data when processing data.

3. Support for Semi-structured and Unstructured Decisions:

- DSS on the other hand is intended for usage in cases where the decision-making process is not highly routinized as in the case of traditional management information systems.

4. Flexibility and Adaptability:

- The system can be applied to any type of decision-making environment and is versatile in the sense that it can be modified in exact conformity with the needs of the users or the organization.

5. What-if Analysis:

- It helps in what-if analysis where the assumptions or the values of the input variable can be varied to determine the impact of change on the result.

6. Timely and Relevant Information:

- DSS thus supplies timely and relevant information that can be used by the decision-makers to respond appropriately to requisite and volatile environments.

7. Support for Group Decision Making:

- Most of the available DSS have group support systems, where more than one person is involved in the decision-making process.

Example:

A DSS helps a manager decide the best pricing strategy by analysing demand and costs.

Give the difference between OLAP and OLTP.

Factor	OLAP	OLTP
Primary Goal	Analysis & Reporting	Transaction Processing
Workload	Complex queries	Short updates
Data Structure	Star/Snowflake schema	Normalized schema (3NF)
Query Type	Read-heavy (SELECT)	Read/Write balance
Response Time	Seconds to minutes	Milliseconds
Data Volume	Historical, massive	Current, limited
Concurrency	Low to medium	Very high
Indexing	Bitmap, composite	Primary, foreign key
Data Updates	Periodic batch	Real-time (ACID)
User Type	Analysts, managers	Clerks, customers

Compare star schema, snowflake schema, star constellation.

Factor	Star Schema	Snowflake Schema	Star Constellation
Normalization	Low (denormalized)	Medium (normalized)	Mixed (multiple facts)
Dimension Tables	Single level	Multiple levels	Shared/Conformed
Query Complexity	Simple joins	Complex joins	Very complex
Storage Space	More redundant	Less redundant	Moderate to high
Query Performance	Fast (fewer joins)	Slower (more joins)	Depends on design

Factor	Star Schema	Snowflake Schema	Star Constellation
Ease of Use	Very easy	Moderate	Difficult
Data Redundancy	High	Low	Medium
Hierarchy Handling	Implicit	Explicit (relational)	Per fact table
Typical Use	Reporting/BI	Data warehousing	Enterprise warehouse
Example	One central Sales fact table with Product , Time , Store denormalized dims	One central Sales fact with Product dim normalized into Brand and Category sub-dims	Multiple fact tables (Sales , Inventory , Returns) sharing Time and Product dimensions

Dimensional Modelling.

Dimensional Modelling is a design technique for data warehouses that structures data into facts and dimensions to make it easy for users to query and analyse. It is widely used in Business Intelligence BI systems.

- Fact Table:
 - Contains quantitative data (measures) such as sales, revenue, or quantity.
 - Linked to dimension tables via foreign keys.
 - Dimension Tables:
 - Contain descriptive attributes (who, what, when, where, how).
 - Examples: Customer, Product, Time, Location.
 - Schemas Used:
 - Star Schema: Central fact table connected directly to dimension tables.
 - Snowflake Schema: Dimension tables normalized into multiple related tables.
 - Advantages:
 - Simplifies complex queries.
 - Improves performance of OLAP operations.
 - Provides intuitive structure for business users.
-

BI Architectures

1. Traditional (Waterfall)

- Uses `ETL` to move structured data into a centralized data warehouse. Data is modelled before analysis. Reliable but slow to adapt.

2. Modern (ELT)

- Extracts and loads raw data into cloud warehouses (e.g., `Snowflake` , `BigQuery`), then transforms it. Scales easily and handles large volumes.

3. Data Lakehouse

- Combines data lake flexibility (raw, unstructured data) with warehouse performance (ACID transactions, fast queries). Reduces data duplication.

4. Decentralized (Data Mesh)

- Organizes data by business domains (e.g., sales, logistics). Each domain owns its data as a product. Scales teams but requires strong governance.

5. Data Fabric

- A metadata-driven, virtual architecture that connects disparate data sources without moving data. Automates integration and discovery.

6. Real-Time (Streaming)

- Processes live data streams using tools like `Kafka` or `Flink` . Delivers instant insights for monitoring, fraud detection, or personalization.